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Deep Bayesian Active Learning with Image Data

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Abstract

This report investigates the integration of deep learning architectures into an active learning framework, focusing on high-dimensional image data. Traditional active learning relies on models that can evaluate prediction uncertainty and learn efficiently from small datasets—requirements that standard deep learning models typically fail to meet. By utilizing stochastic regularization techniques, specifically Monte Carlo (MC) dropout, Bayesian Convolutional Neural Networks (BCNNs) can successfully approximate Bayesian inference and quantify epistemic uncertainty. We review the theoretical foundations of this approach, examining various uncertainty-based acquisition functions such as BALD, Max Entropy, and Variation Ratios. Finally, we summarize the empirical evaluation of these methods on the MNIST and ISIC 2016 datasets, demonstrating how Bayesian active learning significantly reduces the amount of labeled data required to train robust deep architectures. Our code is available at <https://github.com/d4darius/ActiveLearningBCNN>

1 Introduction

The paper describes its intent to develop an active learning framework with the use of deep learning tools. Specifically, it leverages Bayesian Convolutional Neural Networks (BCNNs) to perform image classification on two major datasets: MNIST and ISIC2016 [2].

It is stated that, while traditional machine learning paradigms rely on collecting large quantities of labeled data, a system that is able to learn from small amounts of data and that can choose by itself which data to label would make machine learning more widely applicable. Active learning provides exactly this methodology, by first training the model on an initial dataset and then deciding, through the use of an acquisition function, which points to ask an oracle (such as a human expert) to produce a label for. This set of newly labeled data will then be added to the training set to train a new model, and the process is repeated iteratively.

However, even though active learning has been proven effective over several applications [8], deep learning poses several challenges to its use due to two major factors.

The first aspect concerns the need for such models to be able to learn from small amounts of data. Indeed, recent advances in deep learning are notorious for their dependence on large amounts of data [6], and this constitutes a serious obstacle.

Secondly, many AL acquisition functions rely on model uncertainty over unseen data. Deep learning methods rarely represent such model uncertainty, which further restricts the class of models that can be used within the active learning framework.

Due to the challenges presented by the application of AL for high-dimensional data, various approaches have been used to tackle the issue. In the past, traditional approaches have relied on kernel methods, applying margin-based uncertainty or RBF kernels on raw images or low-dimensional features to retrieve uncertainty about the predicted label [9, 4, 5, 7]. The authors also mention

semi-supervised learning approaches and argue that, since they use unlabeled data to allow the model to learn the underlying data distribution, they form the closest modern literature to active learning of image data.

The main issue related to using kernel methods is that they cannot capture spatial information in the input image. On the contrary, Convolutional Neural Networks (CNNs) are designed to leverage exactly this spatial information and, by placing a set of prior probability distributions over the model's parameters, we obtain their Bayesian equivalent to be used for AL.

Furthermore, in order to perform approximate inference using a Bayesian CNN, stochastic regularization techniques such as dropout are employed. Inference is done by training a model with dropout before every weight layer, and by performing dropout at test time as well to sample from the approximate posterior (these are stochastic forward passes, referred to as MC dropout) [1].

More formally, this dropout procedure is mathematically equivalent to performing approximate variational inference. In this framework, the goal is to find an approximating distribution $q_{\theta}^*(\omega)$ within a tractable family that minimizes the Kullback-Leibler (KL) divergence to the true model posterior $p(\omega|\mathcal{D}_{train})$. Dropout acts as a variational Bayesian approximation where the approximating distribution $q_{\theta}^*(\omega)$ is a mixture of two Gaussians with small variances, with the mean of one of the Gaussians fixed at zero:

$$q_{\theta}^*(\omega) = (1 - p)\mathcal{N}(m, \sigma^2) + p\mathcal{N}(0, \sigma^2)$$

Here, p represents the probability of the Bernoulli dropout mask (i.e., the probability of dropping a unit during the stochastic forward pass). Because the true posterior of a neural network is computationally intractable to evaluate directly, this uncertainty in the weights is instead marginalized over the approximate posterior using Monte Carlo integration.

The retrieval of these uncertainty measurements is particularly relevant because it can be used with acquisition functions. Given a model $\mathcal{M}$, pool data $\mathcal{D}_{pool}$, and inputs $x \in \mathcal{D}_{pool}$, we define an acquisition function $a(x, \mathcal{M})$ as a function of x that the AL system uses to decide where to query next:

$$x^* = \operatorname{argmax}_{x \in \mathcal{D}_{pool}} a(x, \mathcal{M})$$

Because the exact posterior is intractable, we approximate these acquisition functions directly using our approximate distribution $q_{\theta}^*(\omega)$ through Monte Carlo sampling. Since many tasks involving image data are phrased as classification problems, several acquisition functions are available:

- **Max Entropy:** Chooses pool points that maximize the predictive entropy, which represents a general lack of confidence in the prediction. We approximate this by averaging the stochastic forward passes to estimate the predictive probability and computing its entropy.
- **BALD (Bayesian Active Learning by Disagreement):** Chooses pool points that are expected to maximize the information gained about the model parameters. Points that maximize BALD are those where the model is globally uncertain on average, but individual stochastic samples produce highly confident, yet disagreeing, predictions. It is approximated by taking the difference between the entropy of the average prediction and the average of the individual sample entropies.
- **Variation Ratios:** Maximizes $1 - \max_y p(y|x, \mathcal{D}_{train})$, serving as another metric that measures lack of confidence.
- **Mean STD:** An ad-hoc technique that maximizes the mean standard deviation of the prediction probabilities across all classes that x can take.

Finally, the authors propose as a baseline a random acquisition function, which is simply equivalent to choosing a point uniformly at random from the pool.

2 Results

The primary objective of this project was to reproduce the experimental results of the "Deep Bayesian Active Learning with Image Data" paper, focusing on evaluating how effectively Bayesian Convolu-

tional Neural Networks (BCNNs) can leverage uncertainty to perform active learning. The evaluation was divided into two main parts: a standardized image classification task using the MNIST dataset and a more complex, real-world application using the imbalanced ISIC 2016 dataset.

2.1 Comparison of various acquisition functions

For the standard image classification task, the active learning pipeline was initialized with a highly constrained training set consisting of only 20 labeled images from the MNIST dataset, equally distributed as 2 examples per class. To approximate Bayesian inference, the network utilized Monte Carlo (MC) Dropout, executing 10 stochastic forward passes to estimate predictive uncertainty. At each active learning step, the model queried the 10 most informative unlabeled samples from the pool to be added to the training set, iterating this process over 100 acquisition rounds.

At step 0, prior to any active learning acquisitions, all models demonstrated a baseline test accuracy ranging between 53% and 55% and a high initial test loss (between 1.75 and 1.88). To establish a baseline for comparison, we evaluated a uniform Random acquisition strategy.

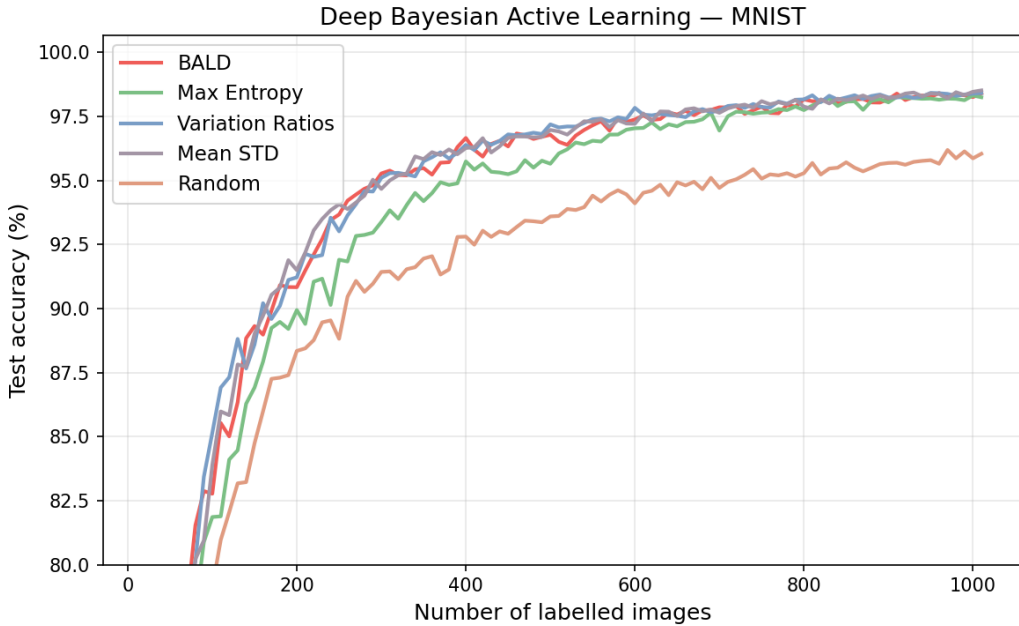


Figure 1: Test accuracy as a function of the number of acquired images, comparing the Bayesian CNN against a deterministic CNN baseline.

In contrast to the Random baseline, the uncertainty-driven acquisition functions (BALD, Max Entropy, Variation Ratios, and Mean STD) demonstrated vastly superior data efficiency and final convergence rates. By systematically querying the most informative samples, these models achieved high performance utilizing significantly less labeled data.

Lastly, as done in the original paper, we report the number of acquisition steps needed to get to test errors of 5% and 10%.

% error	BALD	Var Ratios	Max Entropy	Mean STD	Random
10%	180	220	160	170	260
5%	300	400	300	290	690

Table 1: Number of acquired images to reach a given test error on MNIST.

2.2 Importance of model uncertainty

To assess the impact of modeling epistemic uncertainty during the active learning process, we reproduced the experiment comparing a Bayesian CNN against a deterministic CNN baseline. In our implementation, this evaluation was configured by toggling the network’s behavior at inference time: the Bayesian model utilizes Monte Carlo (MC) dropout to approximate the posterior and capture model uncertainty, whereas the deterministic baseline disables MC dropout at test time, producing a standard point-estimate probability vector.

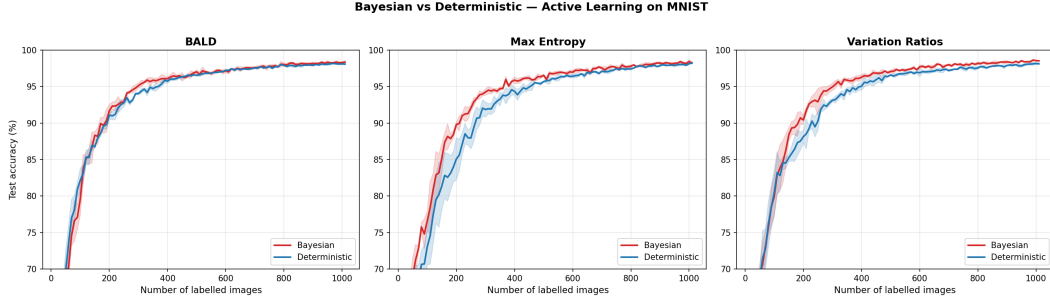


Figure 2: Test accuracy as a function of the number of acquired images, comparing the Bayesian CNN against a deterministic CNN baseline.

As illustrated in Figure 2, we can clearly observe that the test accuracy of the Bayesian CNN consistently lies above the deterministic baseline across the acquisition steps. This behavior is strongly in line with the key findings of the original paper. It demonstrates that capturing epistemic uncertainty allows the model to avoid selecting overconfident yet uninformative samples, leading to a more efficient active learning loop and a higher convergence rate early in the training process.

While our reproduced graph confirms the original conclusions, there are some minor differences in the standard deviations compared to the paper’s results, likely due to the number of experiment repetitions and random initializations. Furthermore, it is worth noting that the corresponding plot in the original paper (Figure 2) displays the accuracy within a restricted range (from 80% to 100%). This specific y-axis scaling visually emphasizes the discrepancy between the two models. Even accounting for these visual and variance differences, the fundamental advantage of the Bayesian approach over the deterministic baseline remains clearly evident.

2.3 Cancer diagnosis from lesion image data

To evaluate the active learning framework on a real-world, unbalanced medical dataset, we reproduced the melanoma diagnosis experiment using the ISIC 2016 dataset [3]. Following the original methodology, our implementation fine-tunes a pre-trained VGG16 model equipped with MC dropout. The active learning loop relies on 20 Monte Carlo samples to estimate model uncertainty, acquiring 100 images per step. To properly isolate the effect of the acquisition functions, the model weights are reset to their pre-trained state before retraining on the newly accumulated datasets.

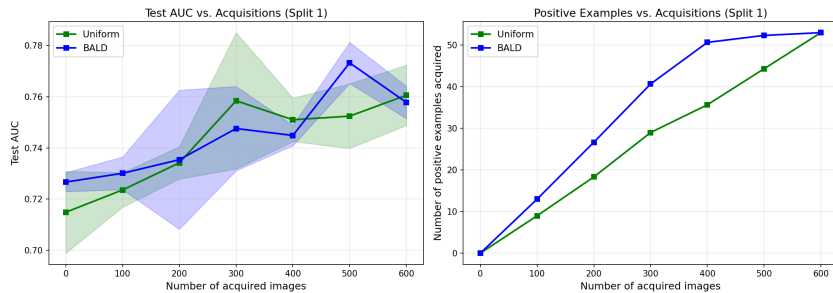


Figure 3: AUC and number of acquired positive examples as a function of acquisition steps for the first random test split on the ISIC dataset.

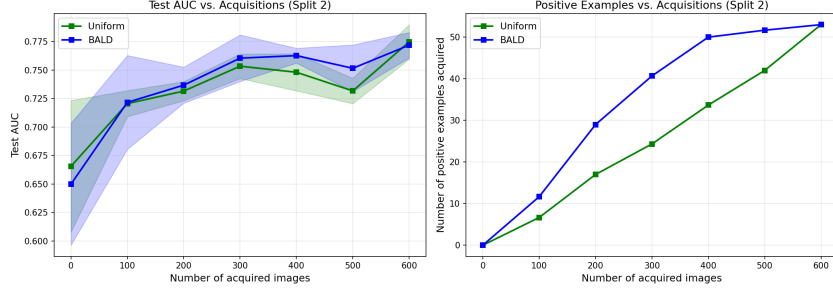


Figure 4: AUC and number of acquired positive examples as a function of acquisition steps for the second random test split on the ISIC dataset.

As illustrated in the performance plots for our experiments (see Figures 3 and 4), we observed some minor discrepancies in the absolute magnitudes of the Area Under the Curve (AUC) and the overall gap between the BALD acquisition function and the uniform baseline when compared to the original paper. However, these variations are fully expected. As noted by the authors of the original study, active learning on small, unbalanced datasets is extremely sensitive to data partitioning. Because the exact random seed from the original experiment is unknown, our random splits naturally produce different test sets and initial pools. Despite this expected variance, our results remain aligned with the paper’s core findings: BALD generally attains higher and more stable AUC values than the uniform baseline.

What remains completely consistent with the original study is the number of positive samples gathered during the acquisition steps. Our graphs clearly demonstrate that BALD acquires a significantly higher number of positive (malignant) examples at each step. This confirms that BALD effectively identifies and queries the most informative, minority-class examples to reduce epistemic uncertainty, whereas the uniform baseline simply selects positive examples at a low rate proportional to their rare frequency in the highly imbalanced pool.

3 Discussion

In general, the original paper was accessible and provided the majority of the necessary details required to reproduce the experimental results.

Creating the Bayesian Convolutional Neural Network (BCNN) required a mechanism to place a prior over the network weights. In our JAX and Flax implementation, this was handled by applying L2 regularization (using weight decay via Optax) during training, which mathematically corresponds to the prior precision in the Bayesian formulation.

The Monte Carlo (MC) dropout was implemented by incorporating `nn.Dropout(rate=..., deterministic=False)` across the layers of the BCNN architecture. To evaluate uncertainty at inference time, we utilized `jax.vmap` to efficiently parallelize the stochastic forward passes across multiple PRNG keys, generating the required MC samples without incurring severe computational overhead.

Notably, we encountered challenges regarding the implementation of the batching structure. The original repository relied on Keras, which natively handles dynamic shapes and incomplete data batches at the end of an epoch. In our JAX implementation, we found that simply leaving a tiny remainder batch (e.g., 2 samples when training on 130 samples with a batch size of 128) produced noisy gradient spikes that led to erratic training behavior. To resolve this without wasting data, we adapted our data loader to merge the incomplete remainder into the final full batch. As a result, the last batch dynamically ranges from the target batch size to at most twice the batch size minus one (e.g., merging a remainder of 2 into a batch of 128 to create a single, clean batch of 130). This solution successfully eliminated the noisy gradients from tiny batches.

Most of the results from the comparison of acquisition functions on the MNIST dataset mirrored the original findings, with only one notable divergence in the Mean STD, where our implementation

outperforms the Random baseline, contrary to Gal et al. (2017) [2], which reported comparable performance between the two. We attribute this discrepancy to the following factors:

- **Dataset variability** - with only 20 initial labeled points (2 per class), the early-stage model is highly sensitive to the specific examples drawn. Mean STD, operating directly on probability scale rather than log-probability scale, is more sensitive to this early-stage miscalibration than information-theoretic measures like BALD or Max Entropy.
- **Number of repetitions** - Both our work and the original paper average over 3 repetitions, which is a small sample. The ranking of Mean STD relative to Random has high variance across seeds, and 3 repetitions may not be sufficient to reliably distinguish them.

Despite this discrepancy, our results confirm the paper’s main finding: BALD, Max Entropy, and Variation Ratios substantially outperform Random and it is worth noting that:

- The Bayesian network converges in a slower fashion and this required an increase in the maximum number of training epochs per acquisition step to reach a performance level comparable to the deterministic model.
- Ultimately, the two networks show comparable overall results, especially when considering the standard deviations of the test metrics. This indicates that the real difference in performance is driven by the sequential learning approach of active learning itself, provided it is combined with a robust acquisition function that efficiently curates the training set.

Lastly, Table 1 presents slight differences from the original paper one, and this is due to the threshold crossing points which are computed from accuracy values averaged across 3 repetitions. This smoothing effect may shift the reported crossings by a small number of labeled examples compared to values derived from individual runs and should be taken into account when comparing the two tables.

Use of Artificial Intelligence

Throughout the development of this project and the drafting of this report, Artificial Intelligence (AI) tools were utilized to assist with both writing and technical implementation. For the documentation, AI was employed to refine raw concepts into polished text and to formalize sentences and academic expressions.

On the technical side, AI was utilized as an assistive tool to generate code and accelerate the debugging process. To ensure the integrity of the work, strict and targeted specifications were consistently provided to the AI to address precise issues. Furthermore, any proposed solutions, code fixes, or implementation plans were heavily revised, critically questioned, and manually validated before being integrated into the final codebase.

4 Workload Distribution

The project workload was subdivided between the two authors as follows:

- **Andrea Gaudino** was responsible for the initial implementation of the Bayesian Convolutional Neural Network (BCNN) and the integration of Monte Carlo (MC) dropout for variational approximation. Additionally, he conducted the experiments comparing the performance of the various active learning acquisition functions.
- **Dario Gosmar** carried out the experimental analysis assessing the importance of model uncertainty by comparing the Bayesian CNN against the deterministic baseline. He was also responsible for extending the Active Learning paradigm to a real-world scenario by designing and conducting the experiments on the ISIC melanoma dataset. Furthermore, he wrote the introduction of this report, summarizing the key findings of the original paper and providing the theoretical background necessary to understand the framework.

The remaining sections of this report, including the results and discussion, were collaboratively authored by both members simultaneously.

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